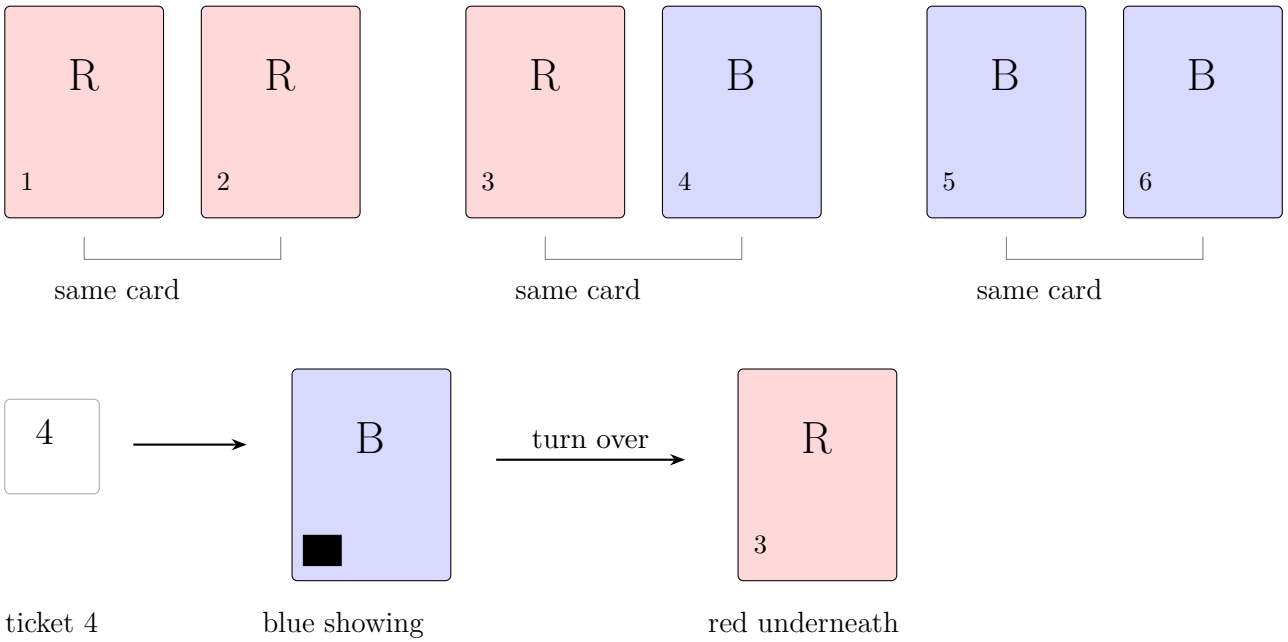
The small numbers match the six tickets. A secret ticket picks the face that shows, with its number covered. You see only the color. Every ticket has the same chance.

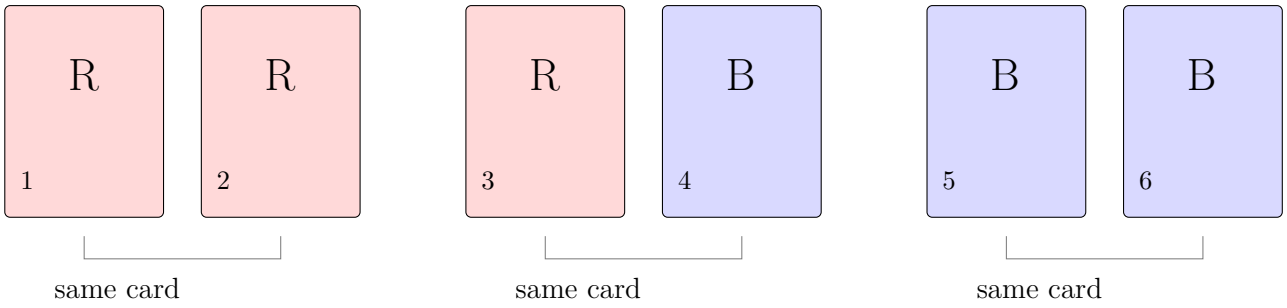


Problem 1: Given only a red showing color, determine the chances of the hidden colors exactly. Do the same for a blue showing color.

red showing

blue showing

Problem 2: A friend says, “Red is showing, so only the red/red and red/blue cards remain. Each is one card, so each must have the same chance.” Does the selection rule support that argument?



Problem 3: The chooser also tells you one fact about the covered number: the number, whether it is odd or even, or whether it is 3 or less. After a red clue, what does each fact do to the better guess?

the number

odd or even

3 or less